Learn LaTeX (Part 1)

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1 Introduction

This is a simple LaTeX document to demonstrate how to use LaTeX for typesetting mathematical equations and creating structured documents.

2 Writing Formulas

Let's begin with a formula $e^{i\pi} + 1 = 0$, which is known as Euler's identity. It beautifully connects five fundamental mathematical constants: e, i, π, 1, and 0.

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

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$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$
2. We can also write matrices:
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$
3. We can also use continuous fractions:
$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}}$$

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3 More Formulas

- Integral:
$$\int_a^b f(x)\,dx = F(b) - F(a)$$
- Triple integral:
$$\iiint_a^b \int_c^d f(x,y)\,dx\,dy$$
- Vector:
$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$
- Limit:
$$\lim_{x \to \infty} \frac{1}{x} = 0$$
- Matrix:
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$